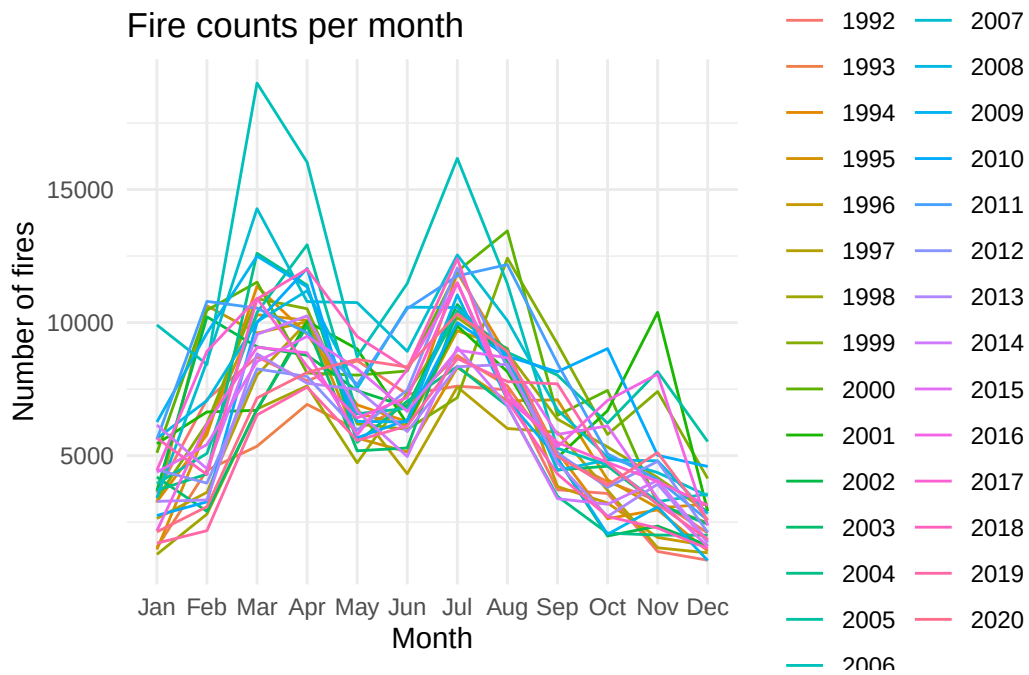


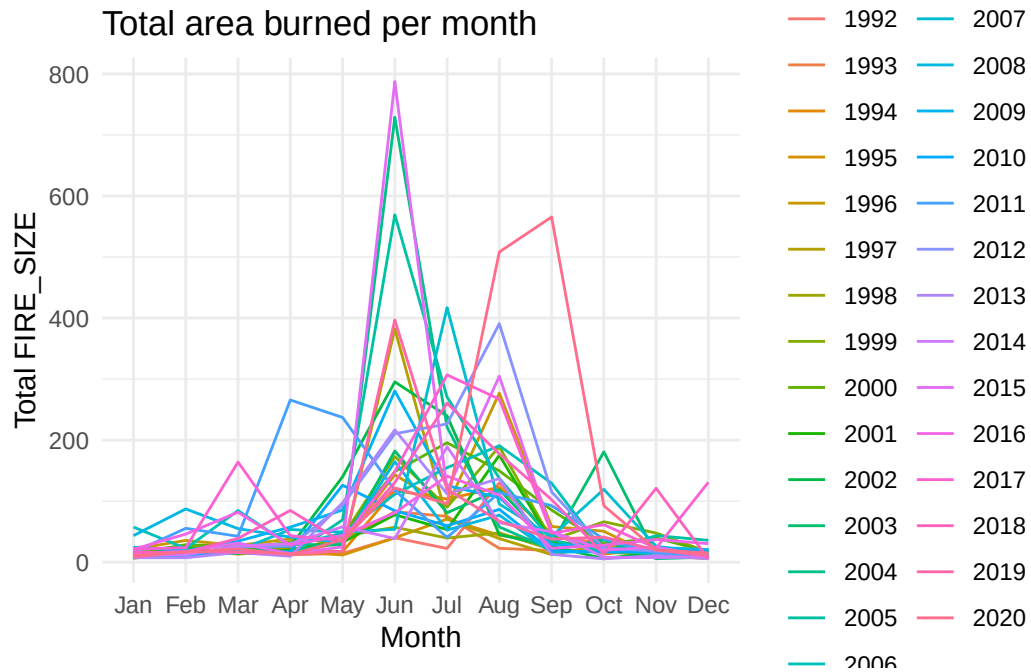
Final Project Report

Emily Dunham & Cooper Maughan

Data Analysis

Exploratory Analysis





Modeling

Regular Model

using seWithMean for s(t.vec) .

$$R^2 = 0.5682962$$

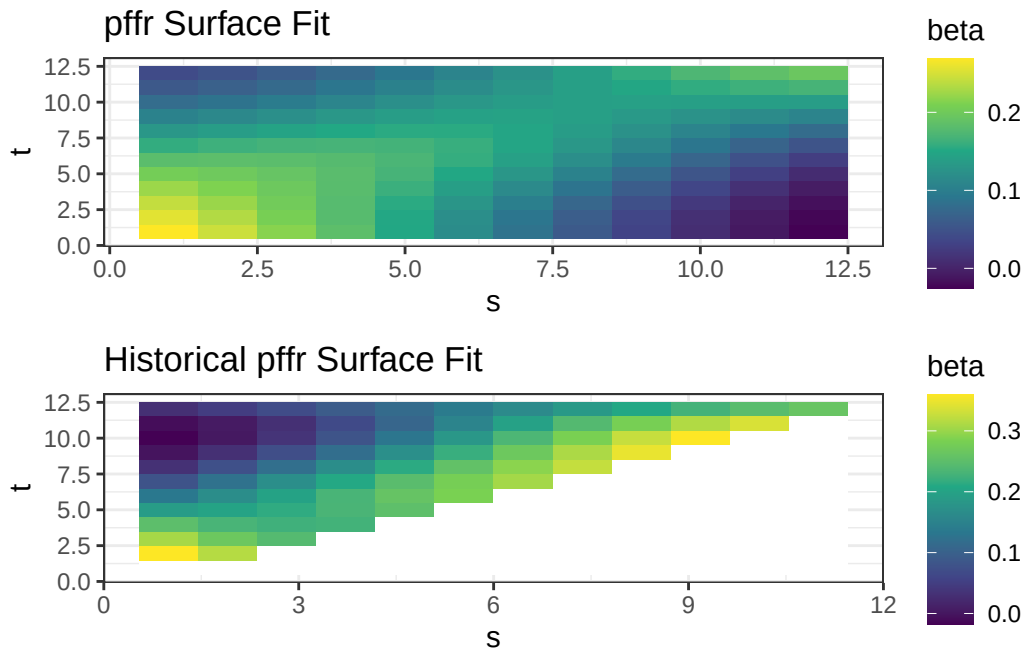
Historical Model

using seWithMean for s(t.vec) .

Plotting the Models

using seWithMean for s(t.vec) .

using seWithMean for s(t.vec) .



Predictions

Full Train Model

using `seWithMean` for `s(t.vec)` .

Historical Train Model

using `seWithMean` for `s(t.vec)` .

Regular Model Prediction

[1] 141.591

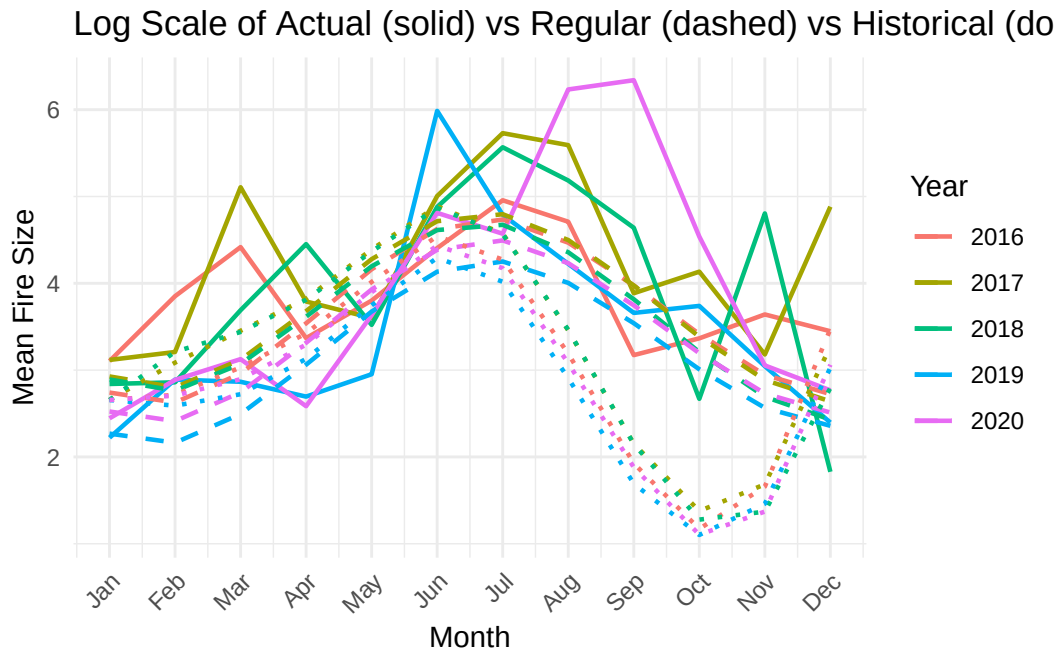
Historical Model Prediction

using `seWithMean` for `s(t.vec)` .

using `seWithMean` for `s(t.vec)` .

[1] 141.9687

Plot Predictions (log scale)



Plot Predictions

```
Train_Model_nl <- pffr(  
  Y_area_train ~ ff(X_fires_train, xind = s),  
  yind = t,  
  method = "NCV",  
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))  
)  
  
Train_FullBetas_nl <- coef(Train_Model_nl, n1 = 12, n2 = 12)$smterms[[2]]$value
```

using seWithMean for s(t.vec) .

```

R2_full_train_nl <- summary(Train_Model_nl)
# R^2 = 0.292

Train_Model_Hist_nl <- pffr(
  Y_area_train ~ ff(X_fires_train, xind = s, limits = 's<t'),
  yind = t,
  method = "NCV",
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))
)

Train_HistBetas_nl <- coef(Train_Model_Hist_nl, n1 = 12, n2 = 12)$smterms[[
  2
]]$value

```

using seWithMean for s(t.vec) .

```

R2_hist_train_nl <- summary(Train_Model_Hist_nl)
# R-squared of .321

reg.train_preds_nl <- predict(
  Train_Model_nl,
  newdata = list(X_fires_train = X_fires_test)
)

# Look at RMSE
y_test <- t(Y_area_test)
reg.train_preds_t_nl <- t(reg.train_preds_nl)

regular_rmse_nl <- sqrt(mean((y_test - reg.train_preds_t_nl)^2))
regular_rmse_nl

```

[1] 104.2018

```

#104.20

# Get intercept
smterms <- coef(Train_Model_Hist_nl, n1 = 12, n2 = 12)$smterms[[1]]$value

```

using seWithMean for s(t.vec) .

```
pterm <- coef(Train_Model_Hist_nl, n1 = 12, n2 = 12)$pterm[1, 1]
```

using seWithMean for s(t.vec) .

```
b0 <- pterm + smterm
b0_matrix <- matrix(b0, nrow = 12, ncol = 5)

# Get the beta surface matrix (12 x 12)
b1 <- Train_HistBetas_nl
b1 <- matrix(b1, nrow = 12, ncol = 12)
b1[lower.tri(b1, diag = TRUE)] <- 0 # taking out values where s>=t

# Manual prediction
b1_matrix <- X_fires_test %*% b1
b1_matrix_t <- t(b1_matrix)

hist.train_preds_nl <- b0_matrix + b1_matrix_t

## Get RMSE's
historical_rmse_nl <- sqrt(mean((y_test - hist.train_preds_nl)^2))
historical_rmse_nl
```

```
[1] 103.5846
```

```
# 103.58

test_years <- 2016:2020

# --- Actual values (long format)
actual_long <- as.data.frame(Y_area_test) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "actual") |>
  mutate(month = as.numeric(month))

# Then redo the long format with back-transformed values
reg_long <- as.data.frame(reg.train_preds_nl) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "pred_reg") |>
  mutate(month = as.numeric(month))
```

```

hist_long <- as.data.frame(t(hist.train_preds_nl)) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "pred_hist") |>
  mutate(month = as.numeric(month))

# Re-join and replot
plot_dat <- actual_long |>
  left_join(reg_long, by = c("year", "month")) |>
  left_join(hist_long, by = c("year", "month")) |>
  mutate(year = factor(year))

ggplot(plot_dat, aes(x = month, color = year, group = year)) +
  geom_line(aes(y = actual), linewidth = 0.8) +
  geom_line(aes(y = pred_reg), linetype = "dashed", linewidth = 0.8) +
  geom_line(aes(y = pred_hist), linetype = "dotted", linewidth = 0.8) +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  labs(
    title = "Actual (solid) vs Regular (dashed) vs Historical (dotted) Predictions",
    x = "Month",
    y = "Mean Fire Size",
    color = "Year"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```

Actual (solid) vs Regular (dashed) vs Historical (dotted) Predic

